



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

29-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : S, I(A), I, LIVE-I & I(A))

Time : 3 Hours

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY
Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

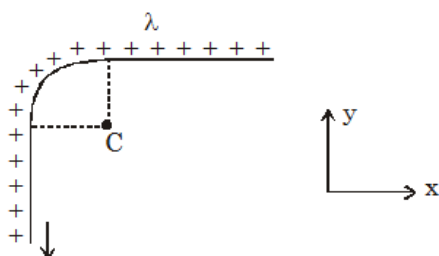
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

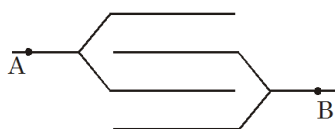
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Two semi-infinite wires are joined with a quarter circular arc as shown in diagram. The charge density is same on both the wires and the arc. x component of electric field intensity at point 'C', the center of arc, is

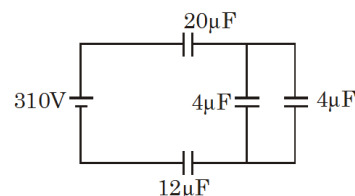


- (A) $\frac{k\lambda}{R}(-\hat{i})$ (B) $\frac{k\lambda}{R}\hat{i}$
 (C) $\frac{2k\lambda}{R}\hat{i}$ (D) $\frac{2k\lambda}{R}(-\hat{i})$
2. There are four plates each of area A (of one side) and separated by a distance 2d from one another. What is capacitance between A & B.

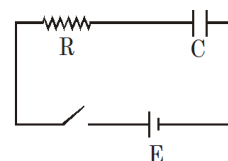


- (A) $\frac{A\epsilon_0}{d}$ (B) $\frac{A\epsilon_0}{2d}$
 (C) $\frac{3}{2} \frac{A\epsilon_0}{d}$ (D) $\frac{2A\epsilon_0}{d}$

3. What is charge and potential difference across the $4\mu\text{F}$ capacitor.

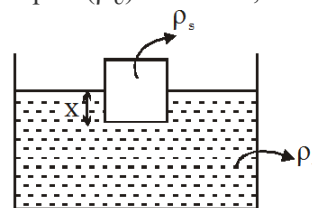


- (A) $640\mu\text{C}$, 160V (B) $280\mu\text{C}$, 70V
 (C) $600\mu\text{C}$, 150V (D) $200\mu\text{C}$, 50V
4. At $t = 0$ switch is closed. Graph between rate of work done (P) by battery wrt time will be



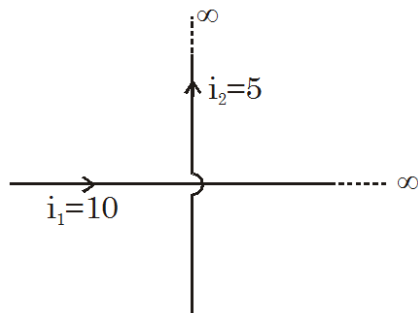
- (A) (B)
 (C) (D)

5. A cube of length ' ℓ ' having density ' ρ_s ' is floating in liquid (ρ_ℓ) as shown, then



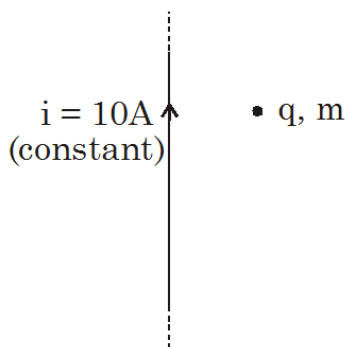
- (A) $x = \frac{\ell\rho_s}{\rho_\ell}$ (B) $x = \frac{\ell\rho_\ell}{\rho_s}$
 (C) $x = \frac{\ell^3\rho_\ell g}{\ell^2 g}$ (D) $x = \frac{\ell^3\rho_s g}{\ell^2 g}$

6. Two long straight conductors with current i_1 & i_2 are placed along x-y axis. Equation of locus of points of zero magnetic field is



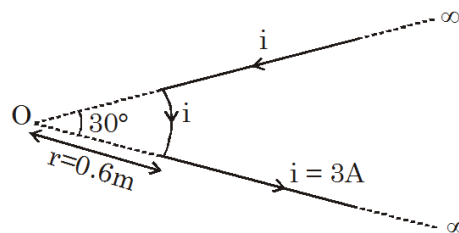
- (A) $y = 2x$ (B) $y = \frac{x}{2}$
 (C) $y = -\frac{x}{2}$ (D) $y = -2x$

7. For the given figure :



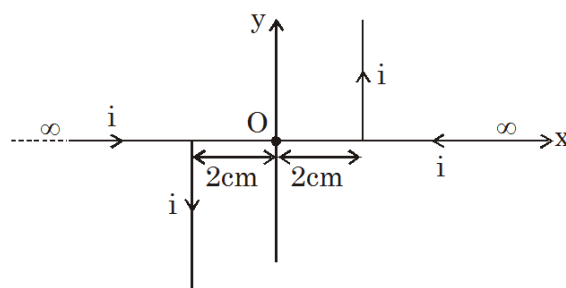
- (A) If charge (q, m) is at rest, it will experience electric force only
 (B) If charge is at rest it will experience magnetic & electric force both
 (C) If charge is moving it will experience both magnetic & electric force
 (D) Charge will not experience electric force whether at rest or in motion.

8. In the given figure B_{net} at point O will be



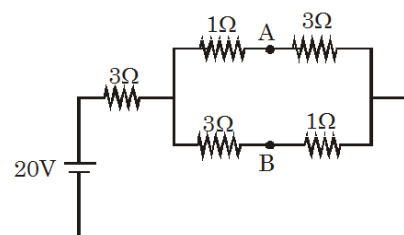
- (A) $1.62 \times 10^{-7} \text{ T}$ (B) $2.62 \times 10^{-7} \text{ T}$
 (C) $3.62 \times 10^{-7} \text{ T}$ (D) $4.62 \times 10^{-7} \text{ T}$

9. In the given figure magnetic field at origin if $i = 20 \text{ A}$ each, will be



- (A) 10^{-4} T (B) $2 \times 10^{-4} \text{ T}$
 (C) $\frac{1}{2} \times 10^{-4} \text{ T}$ (D) $4 \times 10^{-4} \text{ T}$

10. A battery of emf 10V is connected to resistance as shown in figure. The potential difference $V_A - V_B$ between the points A and B is :-



- (A) 4V (B) -2V
 (C) 5V (D) $\frac{20}{11} \text{ V}$

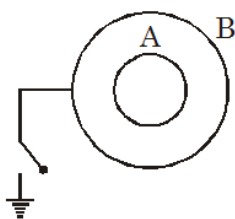
11. An ammeter and a voltmeter are joined in series to a cell. Their readings are A and V respectively. If a resistance is now joined in parallel with the voltmeter,

(A) both A and V will increase
 (B) both A and V will decrease
 (C) A will decrease, V will increase
 (D) A will increase, V will decrease

12. Mark the correct statement :

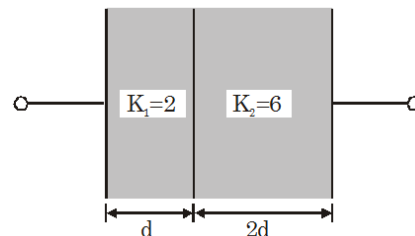
(A) Electric field due to a dipole falls off at large distance as $1/r^2$
 (B) Electric dipole potential falls off, at large distance as $1/r$
 (C) The potential due to a dipole is axially symmetric about p . That is, if you rotate the position vector r about p , keeping θ fixed, the points corresponding to P on the cone so generated will have the same potential as at P .
 (D) Electric field is in the direction in which the potential increases at maximum rate

13. Two concentric shells having radius R and $2R$ have charges q_A and q_B and potentials $2V$ and $(3/2)V$ respectively. Now shell B is earthed and let new charges be q'_A and q'_B . Then $\frac{q'_A}{q'_B}$ will be equal to



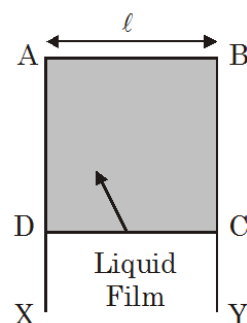
(A) 1 (B) $\frac{1}{2}$ (C) -1 (D) $-\frac{1}{2}$

14. A parallel plate capacitor has two layers of dielectric as shown in figure. If this capacitor is connected across a battery, then the ratio of potential difference across the dielectric layer is :-



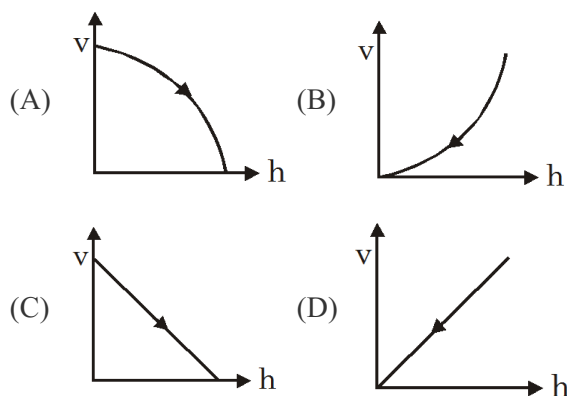
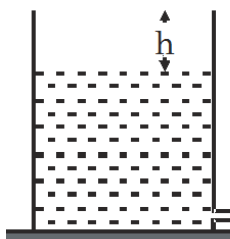
(A) $\frac{4}{3}$ (B) $\frac{1}{2}$
 (C) $\frac{1}{3}$ (D) $\frac{3}{2}$

15. A liquid film is formed over a frame ABCD as shown in figure. A massless wire CD can slide without friction. The mass to be hung from CD to keep it in equilibrium is (Surface tension of liquid is T)

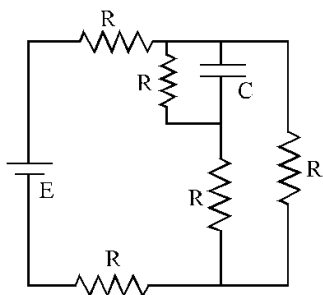


(A) Tl/g (B) $2Tl/g$
 (C) $g/2Tl$ (D) $Tl/2g$

16. A rectangular tank is filled completely with water. A hole at its bottom is unplugged. The graph between the velocity of efflux (through a small hole) vs depth of water h from the top of tank is:

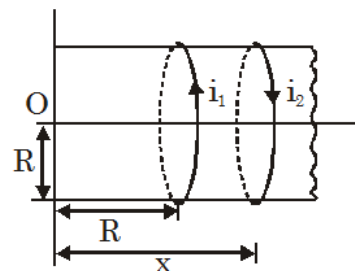


17. Find the time constant of the circuit shown in figure.



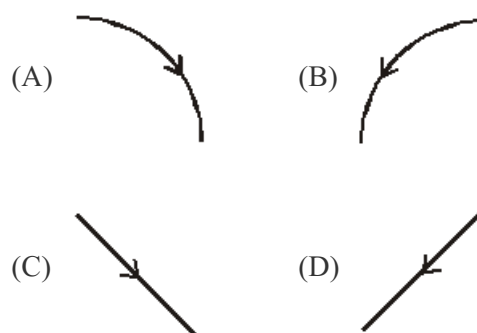
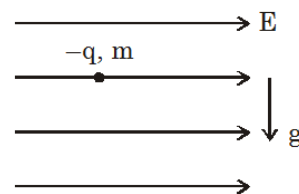
- (A) $2RC/3$
 (B) $5RC/3$
 (C) $5RC/8$
 (D) none of these

18. Two wires are wrapped over a wooden cylinder to form two coaxial loops carrying currents i_1 and i_2 . If $i_2 = 8i_1$ the value of x for $B = 0$ at the origin O is :

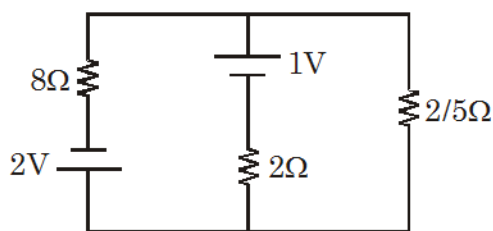


- (A) $\sqrt{(\sqrt{7}-1)}R$ (B) $\sqrt{5}R$
 (C) $\sqrt{3}R$ (D) $\sqrt{7}R$

19. A negatively charged particle is placed in uniform horizontal electric field. Charge particle move under the action of electric force and gravitational force. If initially charge particle is at rest, then which of the following diagram correctly represent the path of the particle :-



20. The current flowing through $\frac{2}{5}\Omega$ resistor will be :-



- (A) $\frac{1}{5}$ A
(B) 1 A
(C) 5 A
(D) $\frac{2}{5}$ A

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

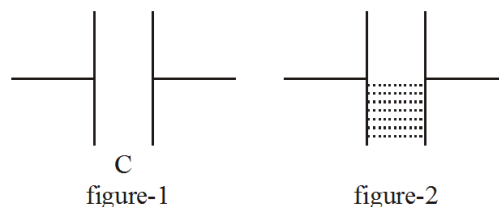
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

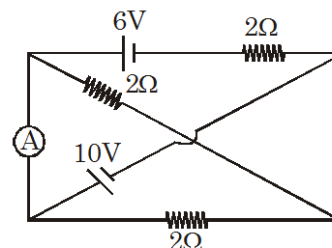
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

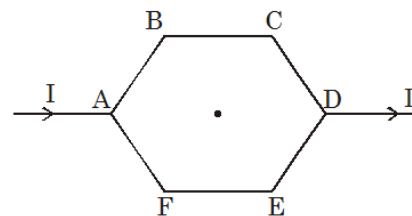
1. A uniform electric field $2\hat{i} - 3\hat{j} + 4\hat{k}$ volt/m exist in space. The electrostatic potential energy stored in a cube of side 1m (in Joules) is $n\epsilon_0$. The value of n is.
2. A parallel plate air capacitor has a capacitance C (figure-1). When it is half filled with a dielectric of dielectric constant 5 (figure-2), the percentage increase in the capacitance will be:-



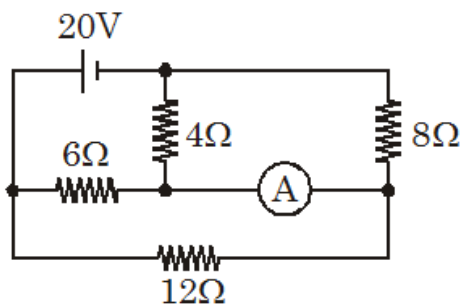
3. An ideal ammeter is connected in a circuit as shown in circuit diagram. What will be the reading of ammeter (in S.I. units) ?



4. ABCDEF is a regular hexagon of side a. Resistance of ABCD is 2 times that of AFED. An electric current I enters at A and leaves at D. The magnetic field at the centre of the hexagon is $\left(\frac{\alpha}{\sqrt{3}} \frac{\mu_0 I}{4\pi a}\right)$ directed out of paper. Find the value of α .

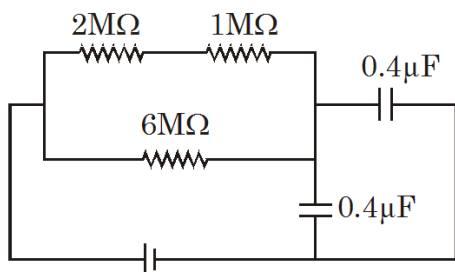


5. The reading of ammeter (in A) is :

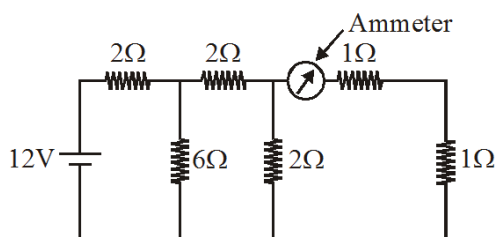


6. Reading of a spring balance when a block is suspended from it in air, is 100. The reading changes to 80 when block is fully submerged in water. Then specific gravity of block must be

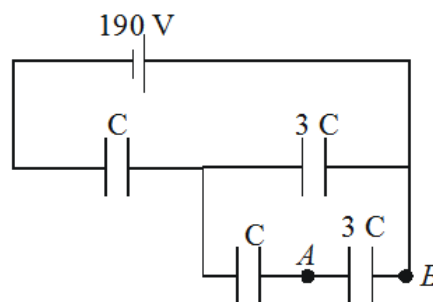
7. Time constant (in sec) of circuit will be



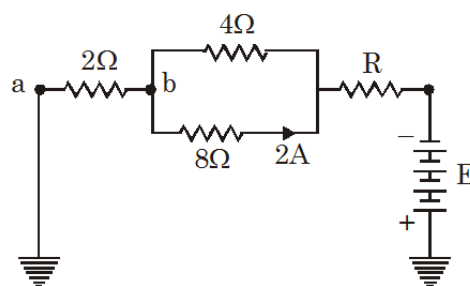
8. In the circuit shown in figure the reading of ideal ammeter (in A) is :-



9. In the circuit shown the potential difference between A and B is x V. Find the value of $\frac{x}{2}$.



10. In the circuit shown, the current in 2Ω resistor (in A) is



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

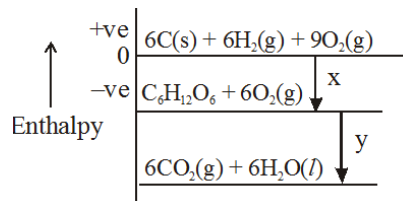
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- An ammonia-ammonium chloride buffer has a pH value of 9 with $[\text{NH}_3] = 0.25\text{M}$. What will be the new pH if 500 ml 0.1 M KOH is added to 200 ml buffer solution ($K_b = 2 \times 10^{-5}$) [$\log 2 = 0.3$]
(A) 8.4 (B) 9.6 (C) 5.6 (D) 4.4
- The self ionization constant for pure formic acid $K_{\text{sic}} = [\text{HCOOH}_2^+][\text{HCOO}^-]$ has been estimated as 10^{-6} M^2 at room temperature. The density of formic acid is 1.15 g/cm^3 . What percentage of formic acid molecule is converted to formate anion:
(A) 0.001 % (B) 0.002%
(C) 0.003% (D) 0.004%
- When one mole of an ideal gas is compressed to half of its initial volume and simultaneously heated to twice its initial temperature, the change in entropy of gas (ΔS) is :
(A) $C_{p,m} \ln 2$
(B) $C_{v,m} \ln 2$
(C) $R \ln 2$
(D) $(C_{v,m} - R) \ln 2$

- Energy level diagram of formation and combustion of components in standard state is given-

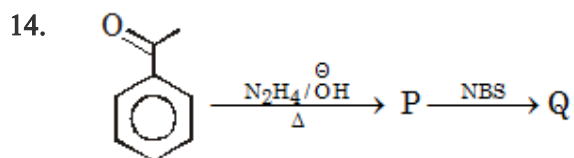


$|X| + |Y|$ is equal to -

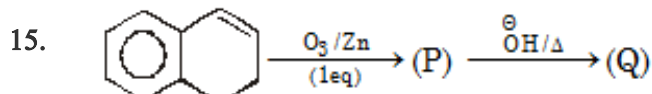
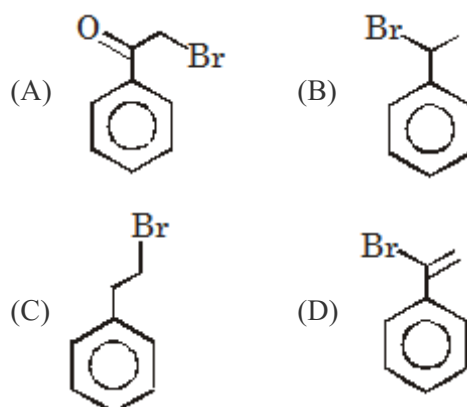
- $|\Delta H_C^0(\text{C(s)})| + |\Delta H_C^0(\text{H}_2)|$
 - $|\Delta H_f^0(\text{CO}_2)| + |\Delta H_f^0(\text{H}_2\text{O})|$
 - $|\Delta H_f^0(\text{glucose})| + |\Delta H_C^0(\text{glucose})|$
 - $|\Delta H_f^0(\text{glucose})| + |\Delta H_f^0(\text{CO}_2)| + |\Delta H_f^0(\text{H}_2\text{O})|$
- When 1 mol of a real gas absorbs 100 kJ heat at constant pressure of 1 bar, it's volume changes from 2L to 2.5 L. What is the enthalpy change for gas -
(A) 150 kJ (B) 50 kJ (C) 100 kJ (D) -50 kJ
 - Two sample of NH_3 each of 10 gm is taken separately in two empty closed rigid vessel of volume 5 litre and 10 litre respectively at 27°C . The degree of dissociation of NH_3 will be
(A) More in the 5 litre vessel
(B) More in the 10 litre vessel
(C) Equal in both vessel
(D) Can't say
 - Assume that the decomposition of HNO_3 is $4\text{HNO}_3(\text{g}) \rightleftharpoons 4\text{NO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$ and the reaction approaches equilibrium at 400K & 30 atm pressure. At equilibrium the partial pressure of HNO_3 is 2 atm. Find K_C at 400 K ($R = 0.08 \text{ l - atm/K-mol}$)
(A) 4 M^3 (B) 8 M^3 (C) 16 M^3 (D) 32 M^3

8. Which of the following reaction is a part of Hall's process :-
- (A) $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
 (B) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$
 (C) $\text{AlN} + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NH}_3$
 (D) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaAlO}_2 + \text{CO}_2 + 2\text{H}_2\text{O}$
9. Which of the following reaction does not take place during steel formation by open hearth process?
- (A) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 (B) $\text{Fe}_2\text{O}_3 + 3\text{Si} \rightarrow 3\text{SiO}_2 + 4\text{Fe}$
 (C) $\text{Fe}_2\text{O}_3 + 3\text{Mn} \rightarrow 3\text{MnO} + 2\text{Fe}$
 (D) $6\text{CaO} + \text{P}_4\text{O}_{10} \rightarrow 2\text{Ca}_3(\text{PO}_4)_2$
10. Which of the following anion(s) produce yellowish white turbidity with Cl_2 water.
- (A) S^{2-} (B) SO_3^{2-} (C) NO_2^- (D) CO_3^{2-}
11. Which of the following will not give positive chromyl chloride test?
- (A) CaCl_2 (B) HgCl_2
 (C) ZnCl_2 (D) $\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$
12. 'A' salt produce white ppt on reaction with $\text{CaCl}_2(\text{aq.})$ and $\text{BaCl}_2(\text{aq.})$ both. Salt may contain following anion :-
- (A) CO_3^{2-}
 (B) SO_3^{2-}
 (C) S^{2-}
 (D) Both (A) and (B)

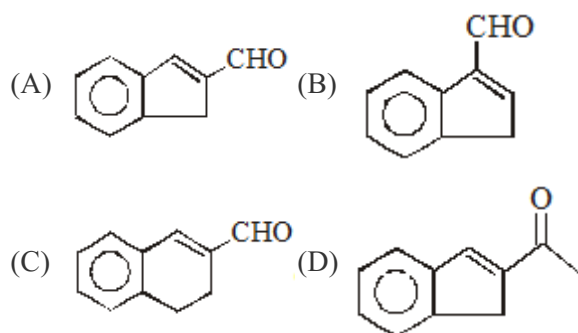
13. Which set of anions gives same colour gas with dilute acid and concentrated acid respectively.
- (A) Br^- , NO_3^- (B) NO_2^- , NO_3^-
 (C) I^- , NO_2^- (D) NO_2^- , Br^-



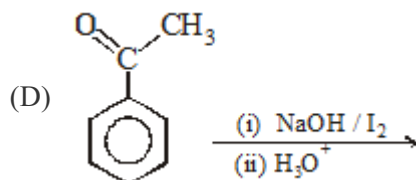
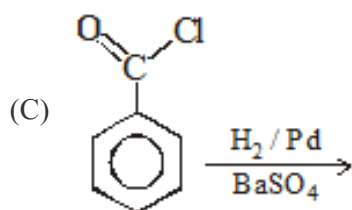
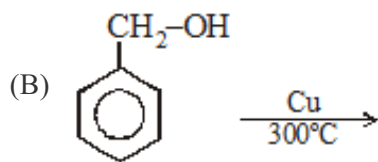
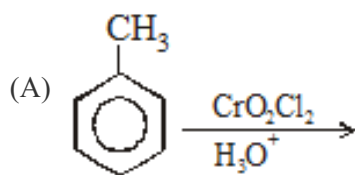
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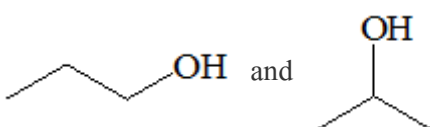


16. Which of the following reaction will not produce benzaldehyde as major product ?



17. Select incorrect statement?

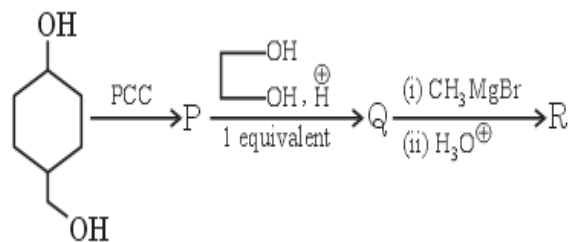
(A) MeCHO and Ph-CHO can be differentiated by Fehling solution

(B)  can be differentiated by Victor Mayor's test.

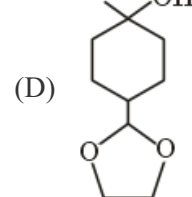
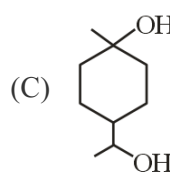
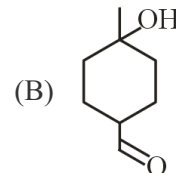
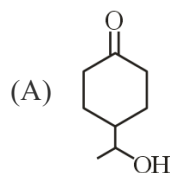
(C) 2-Pentanone and 3-Pentanone can be differentiated by NaHSO₃.

(D) Butanal and butanone can be differentiated by 2,4-DNP.

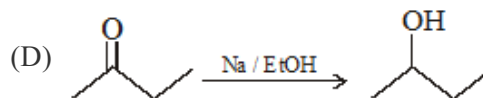
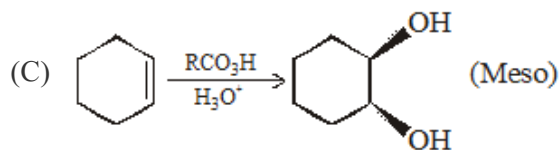
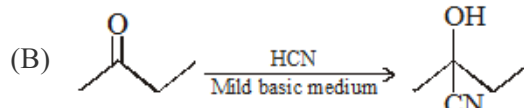
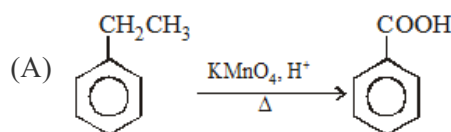
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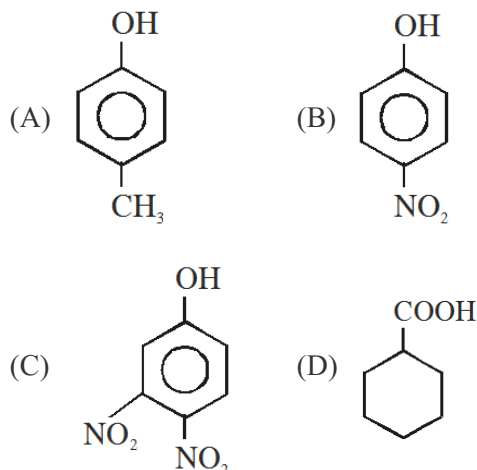
Final product is :



19. Find incorrectly matched reaction with it's major product?



20. Which of following compound doesn't react with NaHCO_3 , giving CO_2 ?



SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. A solution is saturated in SrCO_3 and SrF_2 . The $[\text{CO}_3^{2-}]$ was found to be 10^{-3} mol/L . If the concentration of F^- in solution is represented as $y \times 10^{-2} \text{ M}$ then what is the value of y ?
[Given : $K_{\text{sp}}(\text{SrCO}_3) = 2.5 \times 10^{-10}$; $K_{\text{sp}}(\text{SrF}_2) = 10^{-10}$]

2. The decomposition of hydrogen peroxide follows first order kinetics.



If the volume of O_2 gas liberated at STP in first 20 minutes of the start of decomposition is 25.00 ml, what should be the total volume of O_2 gas (in ml) collected in time, t ($t \gg t_{1/2}$)

(Given : $t_{1/2}$ for the decomposition of H_2O_2 is 10 min)

3. At 500 K for an isobaric process,

$$\Delta S_{\text{system}} = -10 \frac{\text{J}}{\text{mol K}}$$

$$\text{and } \Delta S_{\text{surr}} = 12 \frac{\text{J}}{\text{mol K}}$$

Therefore, ΔG (in kJ/mol) for the entire process will be :-

4. How many of them are sulphide ore ?

Iron Pyrite, Chalcopyrite, Galena, Chalcocite, Magnetite, Cinnabar, Sphalerite, Alumina, Siderite, Calamine, Malachite.

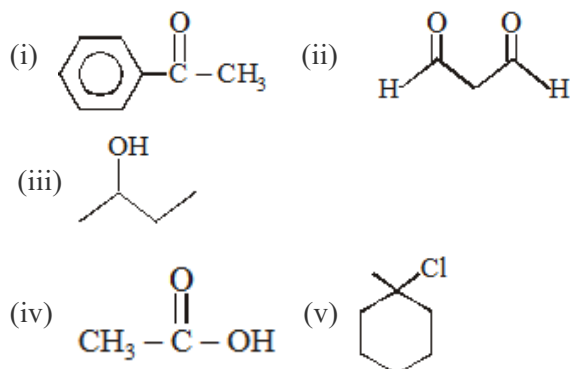
5. Total number of carbonyl ligands which are present as bridging ligand in $\text{Fe}_2(\text{CO})_9$ and $\text{Mn}_2(\text{CO})_{10}$ are x and y respectively, then value of $x + y$?

6. In salt analysis among the following, total number of acidic radicals gives colourless volatile gas/vapour as a observation with dilute H_2SO_4 is.

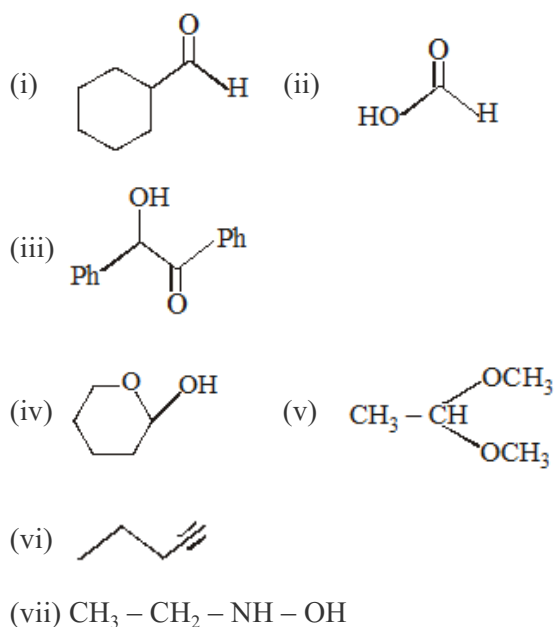


7. Salt of iron $\xrightarrow{\text{Na}_2\text{S}_2\text{O}_3}$ Violet complex $\xrightarrow{\text{On S tan ding}}$ Green solution
(X) (Y) (Z)
Write sum of the oxidation number of Fe in (X), (Y) and (Z).

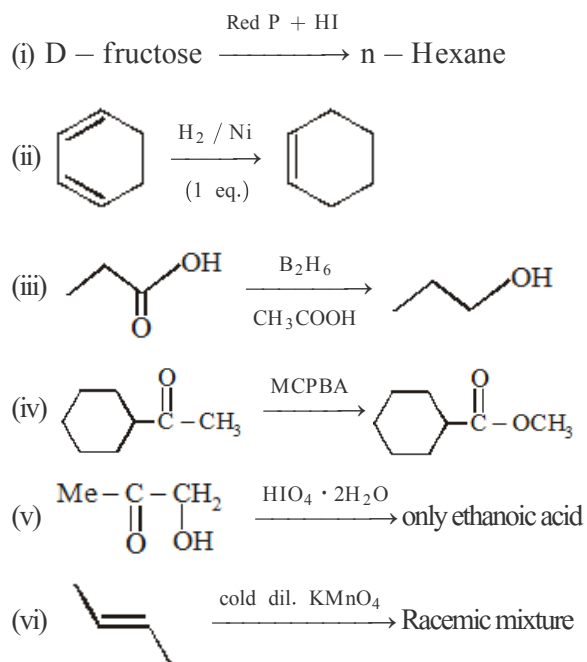
8. How many number of compound(s) giving positive Iodoform test?



9. Find the number of compound giving silver mirror ppt with Tollen's reagent?



10. Find number of reaction giving correct major product?



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ satisfy equation $(A + B)^2 = A^2 + B^2 + 2AB$, then $\frac{a}{b}$ (where $a, b \neq 0$) is equal to
 (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 4
- A square matrix M of order 3 satisfies $M^2 = I - M$, where I is an identity matrix of order 3. If $M^n = 2I - 3M$, then n is equal to -
 (A) 4 (B) 5 (C) 6 (D) 7
- Number of values of $\theta \in [0, 2\pi]$ satisfying the matrix equation

$$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \begin{bmatrix} \cos 2\theta & 0 \\ -\sin 2\theta & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \sin \theta \\ -\frac{\sqrt{3}}{2} & -\cos \theta \end{bmatrix}$$
 is -
 (A) 2 (B) 3 (C) 4 (D) 6
- If curves $ay + bx^3 = 10$ ($a, b \neq 0$) & $8x = y^3$ intersect each other orthogonally at point (1,2), then-
 (A) $a = 4, b = 2$ (B) $a = 2, b = 4$
 (C) $a = 3, b = 4$ (D) $a = 2, b = 6$

- Rolle's theorem holds for the function $x^3 + \alpha x^2 + \beta x + 2$, $0 \leq x \leq 2$ at the point $x = \frac{1}{2}$, then $\alpha + 2\beta$, is equal to-
 (A) $-\frac{3}{4}$ (B) $-\frac{5}{4}$ (C) $\frac{7}{4}$ (D) $\frac{3}{4}$
- The values of 'a' for which the function $f(x) = (a + 2)x^3 - 2ax^2 + 4ax - 1$ decreases monotonically for all real x are-
 (A) $a \in (-\infty, -3] \cup [0, \infty)$
 (B) $a \in (-\infty, -2)$
 (C) $a \in (-\infty, -3]$
 (D) $a \in (-3, 0)$
- Let $y = \left(1 + \frac{1}{x}\right)^{2x}$, then $y'(1)$ is equal to (where $y'(x)$ denotes derivative of y with respect to x) -
 (A) $4(2\ln 2 - 1)$ (B) $4(2\ln 2 + 1)$
 (C) $8\ln 2 - 1$ (D) $8\ln(2e)$
- Consider the function

$$g(x) = \begin{cases} \frac{\max\{x, \frac{1}{x}\}}{\min\{x, \frac{1}{x}\}} & ; \text{ when } x \neq 0 \\ 1 & ; \text{ when } x = 0 \end{cases}$$

 , then
 (A) $\lim_{x \rightarrow 0^+} g(x) = 0$
 (B) $\lim_{x \rightarrow 0^-} g(x) = 1$
 (C) $g(x)$ is continuous for all x except at $x = 0$
 (D) $g(x)$ is differentiable for all x except at $x = 0$

9. Number of ways in which 10 examination papers can be arranged so that the best and the worst papers never come together, is-

- (A) $2(^9C_2)$ (B) $2(^9C_2)8!$
(C) $2(^9C_2)7!$ (D) $8!$

10. The number of ways in which the letters of the word 'LONDON' can be rearranged if the two 'O' s are together but the two 'N's are separated is

- (A) 72 (B) 71 (C) 36 (D) 35

11. There are three parallel lines lying on the same plane. If 6 distinct points are taken on each of these lines, then maximum possible number of triangles with vertices at these points is-

- (A) 756 (B) 396 (C) 306 (D) 486

12. $\int \frac{e^{5\log_e x} - \pi^{\log_e x}}{e^{6\log_e x} + \pi^{2\log_e x}} dx = f(x) + C$, then $f(2) - f\left(\frac{1}{2}\right)$ is -

(Where C is constant of integration)

- (A) 3 (B) 5 (C) 4 (D) 0

13. If $I_1 = \int_1^e \frac{\ln(x+1)}{x^2} dx$, $I_2 = \int_0^1 \frac{e^x}{(x+1)^2} dx$, then -

- (A) $I_1 > 1, I_2 > \ln 2$ (B) $I_1 < 1, I_2 > \ln 2$
(C) $I_1 < 1, I_2 < \ln 2$ (D) $I_1 > 1, I_2 < \ln 2$

14. The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \tan^5(3t) dt}{9x^{12}}$ -

- (A) is equal to $\frac{9}{2}$ (B) is equal to $\frac{27}{2}$
(C) is equal to $\frac{9}{4}$ (D) does not exist

15. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{(2\tan^{-1} e^x - \frac{\pi}{2})}{1+x^2} dx$ is equal to -

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi^2}{4}$ (C) $e^{\frac{\pi^2}{2}}$ (D) 0

16. The area bounded by $f(x) = |\sin x|$, $g(x) = |\cos x|$, $4|x| \leq \pi$, is-

- (A) $\sqrt{2} - 1$ (B) $\sqrt{2} + 1$
(C) $2(\sqrt{2} + 1)$ (D) $2(\sqrt{2} - 1)$

17. If the area enclosed between $y = mx^2$ and $x = ny^2$ is $\frac{1}{3}$ square units, then m, n can be roots of (where m, n are non zero real numbers)

- (A) $2x^2 + 5x + 1 = 0$ (B) $2x^2 + 3x + 2 = 0$
(C) $2x^2 + 3x - 2 = 0$ (D) $2x^2 + 5x - 1 = 0$

18. Differential equation of family of straight lines whose each member has slope equal to twice of its non-zero y-intercept, is -

- (A) $2ydx - (2x + 1)dy = 0$
(B) $2ydy - (2x + 1)dx = 0$
(C) $ydx - (2x + 1)dy = 0$
(D) $ydy - (2x + 1)dx = 0$

19. The solution of differential equation $\frac{dy}{dx}(x^3y^3 + xy) = 1$ is -
(where c is arbitrary constant)

- (A) $\frac{1}{x^2} = 1 - y^2 - ce^{y^2}$
(B) $x^2 = 1 - y^2 - ce^{y^2}$
(C) $\frac{1}{x^2} = 1 - y^2 - ce^{-y^2}$
(D) none of these

20. If A and B are two square matrices of order 3 such that $AB = BA$ and $A^2 + AB = 2B^2$. If $\det(A - B) \neq 0$ and $\det(B) \neq 0$, then $\det(AB^{-1})$ is

(A) -2 (B) 0 (C) 8 (D) -8

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Number of solution(s) of the equation $\cos^{-1}(\cos x) = x^2$ is
- If number of seven digit numbers in which first four digits are in ascending order, last four digits are in descending order and middle digit is a perfect square is λ , then value of $\frac{\lambda}{100}$ is
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial function satisfying $f(x+y) = f(x) + f(y) + 3xy(x+y) - 1 \forall x, y \in \mathbb{R}$ and $f'(0) = 1$, then $\frac{f'(2)}{f(1)}$ is equal to

4. Given $f(x) = \ln \left(\prod_{r=0}^{100} \cos \frac{x}{2^r} \right)$ and $f' \left(\frac{\pi}{4} \right) = -\frac{1}{2^\alpha} \cot \frac{\pi}{2^\beta}$, then the unit's digit of $(1 + |\alpha - \beta|)^{2015}$ is

5. Let the radius and height of right circular cylinder is related as $r^2 + h = 5$. Let λ is maximum volume then $\frac{16\lambda}{25\pi}$ is

6. If $\int \frac{\sin^2 x}{\sec^2 x - \cos^2 x} dx = \lambda x - \frac{1}{\mu} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) + C$ (where C is constant of integration), then $\lambda + \mu$ is equal to

7. If $I_1 = \int_0^{\pi} x \sin x \cdot e^{\cos 4x} dx$ and $I_2 = \int_0^{\pi/2} \cos x \cdot e^{\cos 4x} dx$, then the value of $\left[\frac{I_1}{I_2} \right]$ is (where $[.]$ denotes greatest integer function)

8. If area of the region bounded by $x^2 + y^2 - 6|x| - 6|y| + 9 \leq 0$, $\lambda\pi$, then λ is

9. If the orthogonal trajectory of $y = ax^2$ (where a is arbitrary constant) is $y^{\lambda_1} = -\frac{x^{\lambda_2}}{2} + C$ (where C is arbitrary constant), then $\lambda_1 + \lambda_2$ is

10. A be a square matrix of order 2 with $|A| \neq 0$ such that $|A + |A| \text{adj} A| = 0$, where $\text{adj}(A)$ is adjoint of matrix A, then the value of $|A - |A| \text{adj} A|$ is equal to

Space for Rough Work